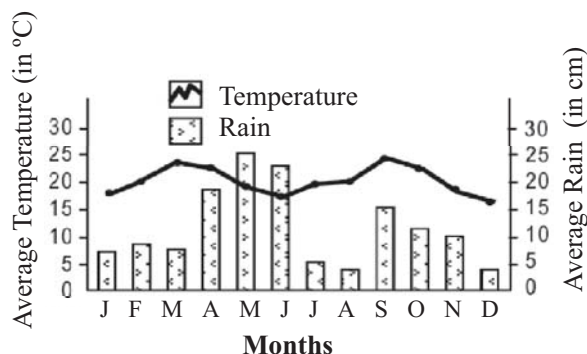


43. The graph given below shows the mean average monthly temperatures (in °C) and mean monthly rainfall (in cm) of a place :



This graph is indicative of which one of the climatic zones of the world ?

- (a) Wet and dry tropical
- (b) Rainy tropical
- (c) Semi-arid tropical
- (d) Temperate marine

I.A.S. (Pre) 1995

Ans. (a)

The given graph indicates a wet and dry tropical zone of the world because the temperature ranges here from 15° to 25°C, with the heaviest rainfall in the months of April, May and June and that also not exceeding 25 cm. Hence, option (a) is the correct option.

44. Hekistotherms are plants growing in –

- (a) bright light
- (b) high temperature
- (c) very low temperature
- (d) very low light

U.P. Lower Sub. (Pre) 2013

Ans. (c)

Hekistotherms are plants which grow in very low-temperature areas like tundra region, mountains and Alpine forests.

45. 'Vegetation is the true index of climate'. This statement is associated with :

- (a) Thornthwaite
- (b) Koppen
- (c) Trewartha
- (d) Stamp

U.P. P.C.S. (Pre) 2018

Ans. (b)

German-Russian botanist and climatologist Vladimir Koppen presented a descriptive classification of the world's climate in the year 1884, which was revised and refined in the years 1918, 1931 and 1936. Koppen has considered vegetation as the true indicator of climate. This has made the five vegetation zones of the world proposed by Candoll in 1874 AD the basis of the division of the world's climate. Later, like Koppen, Thornthwaite also propounded that vegetation is an indicator of climate and that the amount of rainfall and temperature has a substantial effect on vegetation, but he took evaporation into account. On this basis, Thornthwaite has given importance to precipitation effectiveness in demarcation of climate. Thornthwaite has accepted vegetation as an indirect element of climate.

46. In which of the following areas, Mediterranean climate does not prevail?

- (a) Central Chile
- (b) Cape Town
- (c) Adelaide
- (d) Pampas
- (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (d)

The Pampas are fertile South American lowlands that cover more than 750,000 km<sup>2</sup> and include the Argentine province of Buenos Aires, La Pampa, Santa Fe, Entre Rios, Cordoba etc. all of Uruguay and the southernmost Brazilian state, Rio Grande do Sul. The climate is temperate with precipitation of 600 to 1200 mm. The climate is generally temperate, gradually giving way to a more subtropical climate in the north and to a semi-arid climate on the western fringes.

## The Soil

\*Soils are formed as a result of disintegration and decomposition of loosened and unorganized rock materials, fossils of flora and fauna, water and gas. **Bedrock, climatic and biotic factors** play a vital role in the formation of soil. The color, formation and minerals of the soil depend upon the **bedrock**. \***Earthworms** contribute to the formation of the soil and also increase its productivity. Hence, they are known as **engineers of the ecosystem**. Earthworms create pores in the soil, allowing the percolation of air and water

and release of carbon dioxide.

\*The faces of earthworms are helpful in increasing the productivity of the soil. Earthworms soften the soil. They act as decomposers for the soil. \*The excreted urea and dead matter of living organisms (flora and fauna) are decomposed by **decomposers** in the soil.

\*This maintains the nitrogen cycle and keeps providing nitrogen to the soil. \* Soil profile is a vertical section of soil in which all the soil horizons are shown. They are as follows:

(1) Eluviation – The matter of upper stratum descends downwards.

(2) Illuviation – Different salts are introduced from the upper horizon to lower due to percolation of water.

(3) Capillary Action – It is responsible for upward movement of water from wet areas to dry areas of the soil. The smaller the pore the higher the capillary action and vice versa.

\*The capillary function is highest **clay soil** and lowest in **sandy soils**. The effectiveness of capillary action in four different soils is as follows – **Clay > Loam > Sandy soil > Sand**. **Podzol** is a light colored soil of wood regions, mostly found in a cold temperate climate.

\* **Chernozem** soil is found in grassland and transitional areas. It is mostly found in a temperate steppe climate and is also found in humid grasslands. \* **Laterite soil** is found in warm moist areas. The top layer of this soil has **oxides of iron** and salts of aluminum present. \* **Chestnut soil** is found in dry areas. It is dark brown in color and has a low humus content compared to Chernozem soil. \* **Sierozem** soil is found in deserts of mid-latitudes.

\* **Terra Rossa** soil is a reddish silty soil, formed as a result of weathering of limestones and dolomites.

\* **Prairie soil** is found in the grassland areas of temperate regions. The soil is a mixture of Chernozem soil and greyish Podzol soil.

\*Soil conservation is not only about protecting the soil from erosion but also about taking measures to conserve soil fertility. To be precise sustainable use of soil is known as soil conservation.

\* **Soil erosion** is a process through which the top layer of soil along with nutrients is eroded from one place to another.

\* **Contour Bunding** is a method used to conserve soil in hilly areas. During rainfall, it prevents the soil from washing away. Under Contour Banding terrace like structures are made on the slopes of the hilly areas which prevent the soil from

washing away with the downflow of water.

\*Since the tropical forests receive rain throughout the year the soils of the region are suffering from **leaching**. \* **Halophytes** are vegetation that grow well in saline soil.

\*Vegetations growing well in cold regions is called **Cryophytes**. \*Vegetation growing well in rock terrain are called **Lithophytes**, vegetation growing well in tropical deserts are called **Xerophytes** and vegetation growing well in swamps and marshes is called **Hygrophytes**. Hydrophytes are aquatic plants.

**1. Which of the following is the correct order of agricultural practices?**

- (a) Soil preparation - Irrigation - Sowing - Harvesting
- (b) Soil Preparation - Sowing - Irrigation - Harvesting
- (c) Sowing - Harvesting - Irrigation - Soil preparation
- (d) Soil preparation - Irrigation - Harvesting - Sowing

**M.P.P.C.S. (Pre) 2023**

**Ans. (b)**

Agricultural practices are defined as the measures that are followed to raise crops. The process of agriculture includes crop selection, land preparation, seed selection, seed sowing, irrigation, crop growth, fertilizing and harvesting.

**2. The water holding capacity of different types of soils decreases in the order :**

- (a) Clay > Silt > Sand      (b) Clay > Sand > Silt
- (c) Sand > Silt > Clay      (d) Silt > Sand > Clay

**R.O./A.R.O. (Pre) 2017**

**Ans. (a)**

The decreasing order of water holding capacity of the soils given in the option is as follows :  
Clay > Silt > Sand.

**3. Capillaries are most effective in –**

- (a) Clayey soil      (b) Silt soil
- (c) Sandy soil      (d) Loamy soil

**53<sup>rd</sup> to 55<sup>th</sup> B.P.S.C. (Pre) 2011**

**Ans. (a)**

Smaller the pore of the soil, more effective is the capillary activity the bigger the pore of any soil, the less effective its capillary activity will be in clay soil, the capillary activity is most effective and it is least in sandy soil. The effectiveness of the capillary activity in these four soils is as follows :  
Clay > silt soil > loamy soil > sandy soil

4. Conservation of soil is the process in which –

- (a) Barren land is converted into fertile
- (b) Soil is aerated
- (c) Soil is eroded
- (d) Soil is conserved from harm

39<sup>th</sup> B.P.S.C. (Pre) 1994

Ans. (d)

In a broad sense, the meaning of soil conservation is not only controlling erosion, rather it aims at maintaining high-level fertility of the land. The improvement in land use is called soil conservation. Hence, the correct answer is (d).

5. The formal development of Terrarossa takes place in that part of land which consists of

- (a) Limestone
- (b) Cynite
- (c) Granite
- (d) Sandstone

39<sup>th</sup> B.P.S.C. (Pre) 1994

Ans. (a)

Terrarossa is a type of red clay soil produced by the weathering of limestone and dolomites rocks.

6. Soil leaching is a major problem –

- (a) In tropical rainforests
- (b) In African savanna
- (c) In mediterranean regions
- (d) In Australian Dawns

U.P.P.C.S. (Pre) 1998

Ans. (a)

Tropical rainforest regions of the world are facing the soil leaching problem due to heavy rainfall throughout the year. Hence, the correct answer is option (a).

7. Halophytes grow well in –

- (a) Acidic soils
- (b) Cold soils
- (c) soil containing calcium
- (d) Saline soils

M.P.P.C.S. (Pre) 2014

Ans. (d)

Halophytes grow well in saline soil. Cryophytes grow well in cold soil. Plants growing in strong rocks are called lithophytes and xerophytes grow in tropical desert areas. Hygrophytes grow well in waterlogged areas and hydrophytes are aquatic plants.

8. Which of the following adds/nitrogen to the soil?

1. Excretion of urea by animals

2. Burning of coal by man

3. Death of vegetation

Select the correct answer using the codes given below :

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2013

Ans. (c)

Excretion of urea by animals and dead vegetation are decomposed by decomposers. This process helps plants get nutrients from them. In this way, the continuous process of the nitrogen cycle goes on and the soil gets the required amount of nitrogen.

9. Consider the following statements :

Assertion (A) : Of the soil species, clay holds maximum water.

Reason (R) : Porosity is found to be of large size in clay.

- (a) (A) and (R) both are correct, and (R) is the correct explanation of (A).
- (b) (A) is correct, and (R) is wrong.
- (c) (A) and (R) both are correct, but (A) is not the correct explanation of (R).
- (d) (A) is wrong, but (R) is true.

U.P. P.C.S. (Pre) 2015

Ans. (c)

Due to the smaller diameter of the soil particles, it has more water-holding capacity. Hence statement (A) is correct. Due to the small size of the particles of clay, the pore space is found to be of small size, hence reason (R) is wrong.

10. Contour bunding is a method of soil conservation used in –

- (a) Desert margins, liable to strong wind action
- (b) Low flat plains, close to stream courses, liable to flooding
- (c) Scrublands, liable to spread to weed growth
- (d) None of the above

I.A.S. (Pre) 2013

Ans. (d)

Contour bunding method of soil conservation is used in mountain regions. By slowing the flow of water during rain it checks soil erosion. In this process, farming on mountain slopes is done by contour bunding. Contours facilitate easy flow of water thus reducing soil erosion.

11. Match List-I and List-II and select the correct answer by using the codes given below:

| List-I (Soils) | List-II (Climate region)  |
|----------------|---------------------------|
| A. Podzol      | (i) Temperate cold steppe |
| B. Chernozem   | (ii) Cold temperate       |
| C. Spodosols   | (iii) Hot and humid       |
| D. Laterite    | (iv) Humid cold temperate |

Code:

|     | A   | B   | C   | D   |
|-----|-----|-----|-----|-----|
| (a) | ii  | i   | iv  | iii |
| (b) | iii | iv  | ii  | i   |
| (c) | ii  | iii | iv  | i   |
| (d) | iv  | i   | iii | ii  |

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (a)

The correct match of List-I with List-II is as follows :

| (Soils)   |   | (Climate region)      |
|-----------|---|-----------------------|
| Podzol    | — | Cold temperate        |
| Chernozem | — | Temperate cold steppe |
| Spodosols | — | Humid cold temperate  |
| Laterite  | — | Hot and humid         |

12. Assertion (A) : Earthworms are not good for agriculture.

Reason (R) : Earthworms break down the soil into fine particles and make it soft.

Code :

- (a) (A) and (R) both are correct, and (R) is the correct explanation of (A).  
 (b) (A) is correct, and (R) is wrong.  
 (c) (A) and (R) both are correct, but (A) is not the correct explanation of (R).  
 (d) (A) is wrong, but (R) is true.

Jharkhand P.C.S. (Pre) 2013

Ans. (d)

Earthworms give a special contribution to the formation of soil and increase its productivity. They are called the 'ecosystem engineers'. They construct pores in the soil through which water and oxygen enter and carbon dioxide comes out. The wastes of the earthworms are important for the productivity of the soil. They convert the soil into fine particles and make it soft. They act as decomposers. Hence, it is evident that the assertion is wrong and the reason is correct.

13. With reference to agricultural soils, consider the following statements :

1. A high content of organic matter in soil drastically reduces its water holding capacity.
2. Soil does not play any role in the sulphur cycle.
3. Irrigation over a period of time can contribute to the salinization of some agricultural lands.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 3 only  
 (c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (b)

The soil water holding capacity (SWHC) can be used to estimate the maximum amount of water stored in the soil, and reflects the capacity of the soil to provide water for plant growth. The organic matter present in the soil makes the soil particles very dense. Due to this the water-holding capacity of the soil increases. Hence statement (1) is wrong. Sulfur is a major component of proteins, vitamins and amino acids. It is commonly found in the form of sulfate in soil and rocks. Plants take it in the form of sulphate. Sulfates continue to be obtained from the oxidation of  $H_2S$  by the decomposition of dead organisms. Thus sulfur is directly in the cycle. Hence statement (2) is also wrong. Role of soil over a period of time, due to irrigation is that the salt comes to the surface of the land and after evaporation of water, its residues remain on the surface. It can also collect salts in the upper layer of the soil. In this way excessive amounts of salts are formed in the soil, which is called salinization. Hence statement (3) is correct. Thus option (b) will be the correct answer.

14. Which of the following is concerned with soil?

- (a) Edaphic (b) Climatic  
 (c) Biotic (d) Topography

U.P.P.C.S. (Pre) 2018

Ans. (a)

Edaphic means an abiotic factor relating to the physical or chemical composition of the soil found in a particular area.

15. Arrange in chronological order :

1. Patpara formation
2. Khetaurhi formation
3. Baghor formation
4. Sihawal formation